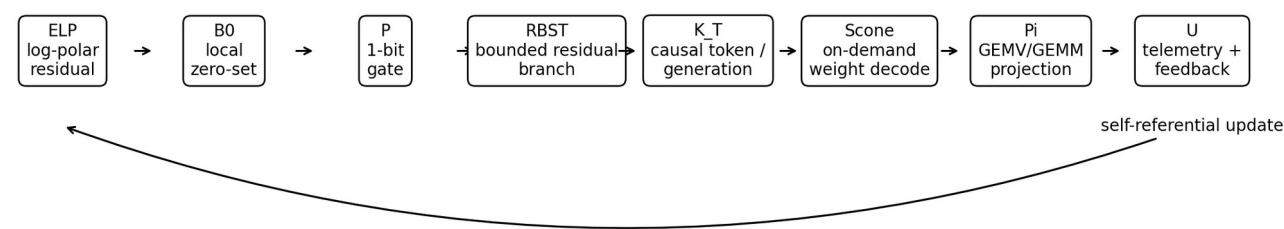


NHDF Edge AI

Qwen3-30B-A3B on a 12 GB RTX 5070 Ti Laptop GPU

Architecture, weight-packing format, edge execution plan, feasibility metrics and falsifiable validation program

NHDF operator chain mapped to the edge weight runtime



The formal NHDF operator chain mapped to an implementable edge-AI weight runtime.

Feasibility	Packed weights	Modeled total VRAM
Conditional / experimental	9.23 GB	11.28 GB at 8K

Prepared for Tom Klootwijk

Version 0.1 | 31 August 2026

Document status

Conceptual research engineering report. The source-derived operator specification is implemented as a semantics-first prototype. Model quality and GPU speed have not been measured on the target laptop; all forward-looking performance figures are explicitly labeled analytical.

Document control and reading guide

Item	Value
Design basis	NHDF Formal Specification v0.1 by Tom Klootwijk [S1], plus the supplied prompt-chain source [S0].
Selected model	Qwen/Qwen3-30B-A3B-Instruct-2507 [M1].
Target	GeForce RTX 5070 Ti Laptop GPU, 12 GB GDDR7 capacity class [H1].
Implemented evidence	Packer, local zero-set projection, log-polar branch score, parity, CRC, selected-row decode, Qwen3 expert adapter, CPU tests, analytical metrics.
Not yet measured	Complete checkpoint size, language-model quality, target CUDA kernel correctness/performance, prefill performance, power and thermal behavior.
ZIP boundary	No upstream model weights are included. Pull and conversion happen locally.

Contents

Section	Purpose
1. Executive conclusion	Decision, fit, expected use and principal risks.
2. Evidence boundary	Separates source requirements, established facts, implementation, projections and hypotheses.
3. Model and hardware selection	Why this model is useful and why it does not fit conventional 12 GB deployment.
4. Formal-to-runtime architecture	Exact mapping of ELP, B0, P, RBST, K_T, Scone, Pi and U.
5. Weight format and distillation	Low-bit codes, local zero sets, residual branches, calibration and integrity.
6. Edge execution	Decode, prefill, Qwen3-MoE integration and memory schedule.
7. Compression and VRAM metrics	Analytical model and scaling with residual budget/context.
8. Speed model	Traffic roofline, sensitivity and what must be benchmarked.
9. Quality and validation	Synthetic smoke result, ablations, fault tests and deployment gates.
10. Applications and operational guide	Practical edge uses and commands.
Appendices	Equations, telemetry contract, package map and references.

1. Executive conclusion

Feasibility verdict: conditional

The proposed mixed-precision NHDF Edge pack is analytically 9.23 GB and the full default 8K profile is 11.28 GB, leaving 0.72 GB of nominal 12 GB capacity. This is enough to justify a build-and-benchmark experiment, but not enough to claim deployment before the actual laptop reports free VRAM, the complete pack is serialized, and the CUDA path is measured.

Qwen3-30B-A3B-Instruct-2507 is the practical stress target because it combines a large 30.5B total parameter footprint with only 3.3B active parameters per token. Its 128-expert, top-8 routing makes expert-selective 2-bit compression and on-demand expert projection technically meaningful [M1][M2]. The upstream BF16 repository is 61.1 GB and the official GPTQ-Int4 checkpoint is 16.9 GB, so neither is an all-weights-resident 12 GB solution [M1][M3].

The implementation does not attempt to make memory literally geometric or to make a GPU execute topology directly. It translates the formal specification into a concrete data representation: each 256-weight group has a non-degenerate local zero-set constraint, residual error is log-polar encoded, a bounded one-bit correction branch is allocated by deterministic error/phase ranking, one-bit parity is a fast event, CRC32 protects each tensor file, and fused kernels reconstruct only the rows needed for a projection.

- Default experts: 2-bit base plus a one-bit sign residual on 15% of groups; modeled 2.321 bits/weight including metadata.
- Attention, embeddings and LM head: groupwise 4-bit; router and normalization weights remain FP16.
- Default context: 8,192 tokens with a modeled int8 KV cache of 0.403 GB.
- Decode traffic: 1.182 GB of packed active weights per generated token; modeled 17.0, 28.4 and 39.8 tokens/s at 3%, 5% and 7% of published bandwidth.
- Production requirement: fused low-bit GEMV for decode, packed GEMM or bounded layer-local dequantization for prefill, and exact CPU/GPU equivalence tests.

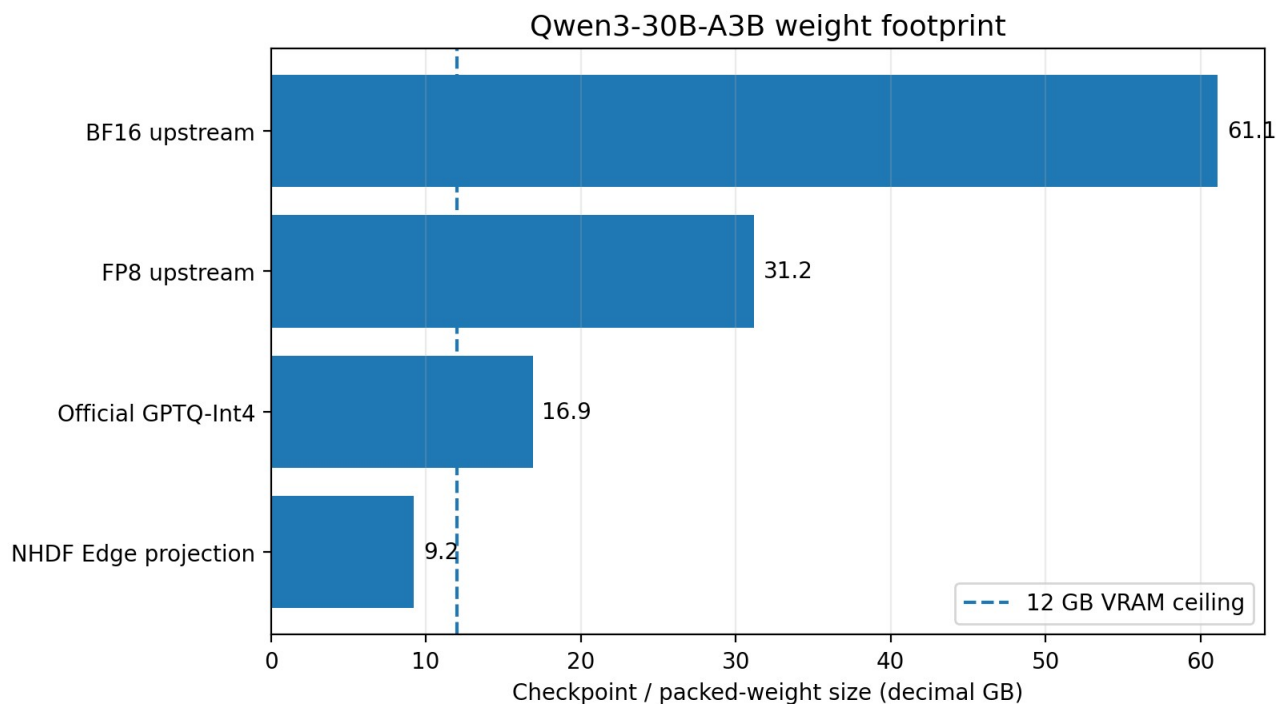


Figure 1. Upstream and projected weight footprints. The 12 GB line is a capacity ceiling, not usable free VRAM.

2. Evidence boundary and corrections to the source narrative

The formal specification already distinguishes its normative core from speculative applications and unsupported hardware claims [S1]. This report maintains that separation. The earlier source transcript is used for vocabulary and traceability, but statements such as mathematical impossibility of crashes, instantaneous processing, storm immunity, direct Tensor Core parity, or near-zero processing cost are not adopted as engineering facts [S0].

Class	What this report treats as evidence	Examples
Source-derived requirement	Normative design intent from the supplied specification.	Operator order, local zero sets, parity limits, bounded branches, causal timeline, feedback.
Established external fact	Current official model, framework or hardware documentation.	Qwen dimensions and file sizes; GPU memory/bandwidth/TGP; official expert tensor layout.
Implemented behavior	Code executed in this package's CPU environment.	Bit packing, reconstruction, local field residual, parity, CRC, selected rows, expert adapter, 41 tests.
Analytical projection	Formula driven by declared assumptions; not a measurement.	9.23 GB weights, 11.28 GB VRAM, bandwidth-based token-rate sensitivity.
Unverified hypothesis	Requires full model and target hardware experiments.	Quality retention, target kernel efficiency, useful speed, thermal stability, application advantage.

2.1 Required non-degenerate reading of SDF zero

A single global field $F(x) = 0$ for every x has zero gradient wherever differentiable. It cannot supply a normal, distance, inside/outside distinction or restoration direction. The formal specification therefore requires local constraints $F_i(z_i, t) = 0$, one per valid cell [S1]. In this weight format, cell i is a quantization group and its local field is the activation-weighted mean reconstruction residual.

$$F_i = \sum_j h_{ij} * (w_{ij} - \hat{w}_{ij}) / \sum_j h_{ij}$$

Choose μ_i so that $F_i = 0$ for every packed group i .

2.2 Parity is an event, not error correction

XOR parity detects an odd number of bit inversions and misses every even number. It neither locates nor repairs damage. NHDF Edge stores the one-bit gate for source conformance and fast diagnostics, while CRC32 protects the complete safetensors file. A deployable corruption-recovery claim would require ECC, redundancy or recomputation [S1].

Safety boundary

The runtime must fail closed on CRC mismatch, missing tensors, impossible geometry, insufficient free VRAM or unresolved meta parameters. Topology is not hardware resilience.

3. Model and target selection

3.1 Why Qwen3-30B-A3B-Instruct-2507

Property	Value	Architectural consequence
Total / active parameters	30.5B / 3.3B	Large stored state, much smaller conditional compute per token.
Layers	48	All-layer weight residency matters; layer offload would penalize every token.
Experts	128 total, 8 active/token	2-bit expert compression yields most of the size reduction; selected expert rows can be projected.
Expert intermediate	768	Each expert has fused gate/up and down matrices with predictable row intervals.
Hidden size	2,048	Group size 256 divides the primary input width exactly.
Attention	32 query heads, 4 KV heads, head dimension 128	Grouped-query attention keeps the KV cache relatively small.
Context	262,144 native maximum	The laptop profile intentionally caps default deployment at 8K; native maximum is not a 12 GB target.

Hugging Face's official Qwen3-MoE implementation stores each expert collection as two 3-D parameters: `gate_up_proj[num_experts, 2*intermediate, hidden]` and `down_proj[num_experts, hidden, intermediate]`. Its reference forward groups tokens by selected expert, performs the two projections and accumulates weighted outputs [M2]. The included adapter preserves this contract rather than flattening experts into unrelated linear modules.

3.2 Target GPU facts and practical constraints

Target fact	Published value	Use in this analysis
VRAM	12 GB GDDR7	Hard nominal capacity; actual free memory must be measured after display/driver allocation.
Memory interface / bandwidth	192-bit / 672 GB/s	Roofline input for decode traffic sensitivity, never treated as sustained application bandwidth.
CUDA cores	5,888	General compute context; not used to infer token rate.
AI throughput	992 AI TOPS	Not used for the low-bit GEMV estimate because format/kernel utilization is unknown.
Power range	60-115 W	Laptop configuration and thermals can materially change sustained performance.

Target fact	Published value	Use in this analysis
Architecture	Blackwell; fifth-generation Tensor Cores	Use only through supported CUDA/PyTorch interfaces. No direct topology or parity execution claim.

The implementation deliberately detects the runtime compute capability instead of baking a speculative PTX sequence into the format. CUDA 12.8 introduced full Blackwell toolkit support, but the exact compiler, PyTorch build, driver and architecture spelling on the laptop must be checked together [H2].

4. Formal-to-runtime architecture

NHDF operator chain mapped to the edge weight runtime

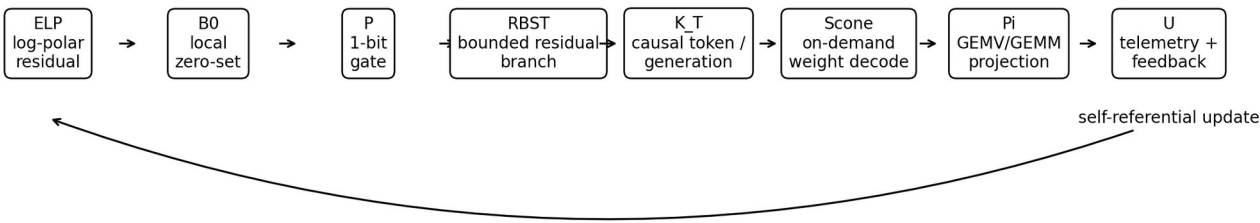


Figure 2. Normative order retained in the weight-runtime mapping.

Operator	Formal role [S1]	NHDF Edge realization	Testable invariant
ELP	Jitter to log-polar address	Residual L2 magnitude to rho; deterministic sine/cosine projections to theta; one-byte bin for selected groups.	Address is deterministic and bounded to declared bins.
B0	Project a candidate to a local zero-level set	Solve one FP16 group mean so the weighted residual average is zero.	$\max F_i \leq \text{declared tolerance.}$
P	One-bit data/topology event	XOR parity of base and residual payload bytes; orientation is a separate diagnostic; file CRC is authoritative.	Stored parity exactly recomputes; even-fault blind spot documented.
RBST	Bounded branch-node bifurcation	Stable sort of group error times phase-curvature factor; allocate exactly the highest-scoring residual budget.	Selected count $\leq$ finite budget; deterministic ties.
K_T	Forward causal propagation	Monotonic tensor conversion order, manifest generation and autoregressive token/cache sequence.	No future-generation read; bounded buffers carry generation tags where used.
Scone	Analytic sweep / extrusion	Decode low-bit codes plus local and residual terms only for requested weight rows or tiles.	No full dense weight is required for batch-one projection.
Pi	Observable projection	Embedding gather, GEMV/GEMM, expert projection and logits.	CPU/GPU outputs agree within declared numerical tolerance.
U	Self-referential state update	Manifest/telemetry, residual metrics, optional Hessian calibration and next conversion/runtime state.	Run emits traceable configuration and failure state.

4.1 What is self-referential here

The current packed state is both operand and update target: the residual produced by reconstructing a source tensor determines log-polar address, parity, branch allocation and the serialized next state. At runtime, outputs and residual telemetry can select recalibration or a new residual budget. This is a bounded closed state transition, not unrestricted recursive memory growth.

$$L_{(n+1)} = U(L_n, \text{Pi}(\text{Scone}(\text{K_T}(\text{RBST}(\text{P}(\text{B0}(\text{ELP}(x_n; L_n)))))))$$

5. Weight representation and reconstruction distillation

5.1 Tensor policies

Tensor class	Default	Reason
MoE expert gate/up and down	2-bit base + 15% one-bit residual	Experts dominate stored parameters and published MoE research supports investigating ultra-low-bit expert-only compression [Q3].
Attention matrices	4-bit groupwise	Attention is active on every token and more sensitive than dormant experts; 4-bit reduces risk.
Token embedding and LM head	4-bit groupwise	Large matrices, but vocabulary/logit quality is sensitive; kept above 2-bit.
Router	FP16	Small and directly controls expert selection.
RMSNorm and other 1-D state	FP16	Negligible capacity gain from lower precision; preserves scaling stability.
KV cache	Modeled int8	Separate runtime choice; must be validated against a concrete cache implementation.

5.2 Group reconstruction

For group i and element j , the reconstructed weight is a local mean plus a low-bit level and an optional sign residual. The mean is solved after the residual branch is chosen, so the local field remains at zero within floating-point tolerance.

```
w_hat_ij = mu_i + s_i * level(q_ij) + m_i * r_i * sign_ij
q_ij: 2-bit or 4-bit code
m_i: residual-mask bit for group i
mu_i, s_i, r_i: FP16 local parameters
```

The converter uses alternating least squares for the base scale and mean. With an optional per-input-channel second moment h_j , both the local field and branch error become activation weighted. This is post-training reconstruction distillation: the source weight/teacher activations define the loss, but the package does not claim full logit-level knowledge distillation.

5.3 Log-polar address and phase-accelerated branch score

```
rho_i = log(1 + gamma * ||residual_i||_2)
theta_i = atan2(<residual_i, sine_basis>, <residual_i, cosine_basis>)
score_i = weighted_MSE_i * (1 + k_phi * |Delta2(theta)_i|)
```

A stable descending sort selects the exact finite branch budget. The golden-ratio concept remains available as a scheduling or geometric scale in the general NHDF specification, but it is not inserted as an unexplained constant into this weight codec. Equal-budget ablations must determine whether any phi-based schedule adds value.

5.4 Storage layout and integrity

Component	Storage	Expert contribution
Base codes	2 bits/weight	2.000 bpp
Residual signs	1 bit/weight on selected fraction f	0.150 bpp at f=0.15
Local mean + scale	2 FP16 values per 256 weights	0.125 bpp
Residual scale	1 FP16 per selected group	0.0094 bpp
Log-polar address	1 byte per selected group	0.0047 bpp
Mask, rank prefix, parity	three 32-bit words per 32 groups	0.0117 bpp
Header/alignment allowance	analytical allowance	0.020 bpp
Total	modeled	2.3208 bpp

Each packed tensor is one safetensors file. The manifest records original shape, group geometry, policy, reconstruction statistics, per-file size and CRC32. Per-group parity supports local diagnostics; CRC32 catches the even-count faults that parity can miss. The complete converted model is not included in the ZIP.

6. Edge execution architecture

6.1 Decode path

Autoregressive batch-one decode is the first optimization target because each token performs matrix-vector products and rereads active weights. The supplied CUDA extension decodes 2-bit/4-bit codes, loads local mean/scale, uses a 32-group mask plus prefix-popcount to locate residual signs/scales, and accumulates directly into the output row. It supports contiguous row intervals so one selected expert can be projected without reconstructing all 128 experts.

```
for output row r in requested interval:
    for input column c in parallel:
        group = r * groups_per_row + c / 256
        w = mean[group] + scale[group] * level(code[r,c])
        if residual_mask[group]:
            rank = prefix[word] + popcount(bits before group)
            w += residual_scale[rank] * residual_sign[rank,c]
        accumulator += x[c] * w
```

Kernel status

The CUDA source is included but could not be compiled in this CPU-only artifact environment. It is a clear experimental GEMV, not a tuned Blackwell kernel. Target compilation, Compute Sanitizer, numerical equivalence and Nsight profiling are mandatory.

6.2 Prefill path

Prompt prefill is matrix-matrix work. Reusing a batch-one GEMV for every prompt token would reread weights and waste available matrix hardware. Version 0.1 therefore treats prefill optimization as an explicit next gate. Two bounded alternatives are proposed: a fused packed GEMM, or layer-local dequantization into the 0.75 GB workspace followed by cuBLAS GEMM. Hidden allocations are prohibited because they invalidate the fit model.

6.3 Full Qwen3 integration

- Build the Hugging Face causal LM on the meta device, so no dense BF16 allocation occurs.
- Replace every ordinary Linear and Embedding with NHDF packed modules.
- Replace each Qwen3MoeExperts object with a paired packed gate_up/down adapter that preserves official router inputs and weighted accumulation.
- Load router and normalization parameters as FP16.
- Reject the model if any meta parameter remains or a pack entry is unconsumed.
- Keep Transformers-specific integration isolated so version drift does not change the serialized format.

6.4 Memory schedule

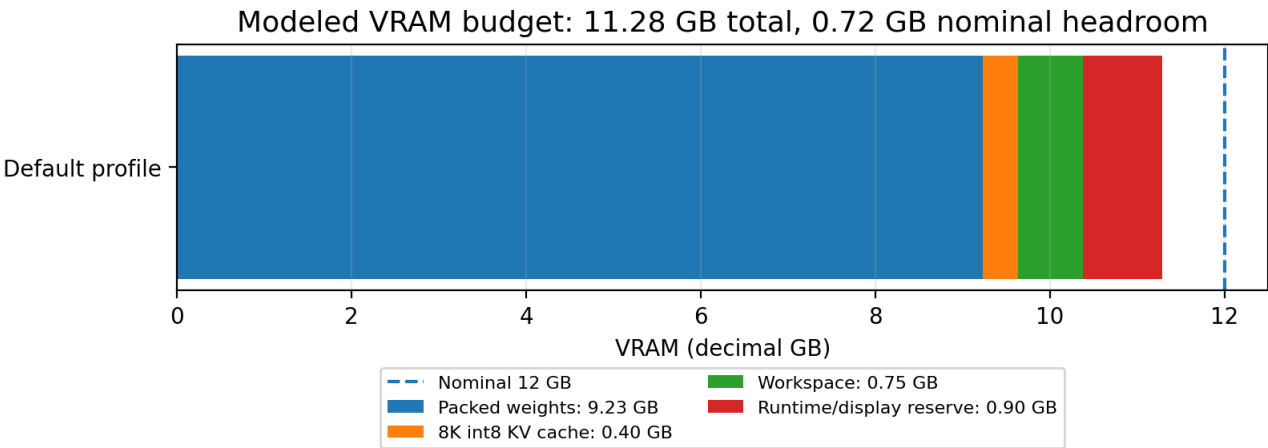


Figure 3. Default analytical VRAM budget. Actual free VRAM, allocator fragmentation and temporary buffers must be measured.

7. Compression and VRAM metrics

Metric status

Every number in Sections 7 and 8 is formula-derived from official model/hardware data and declared format assumptions. The only executed quality result is the small synthetic smoke experiment in Section 9.

7.1 Default result

Metric	Value	Interpretation
Expert effective precision	2.3208 bpp	Includes base, residual and modeled metadata.
Sensitive matrix precision	4.1489 bpp	4-bit code plus mean/scale/parity and allowance.
Packed weights	9.228 GB / 8.595 GiB	Expected serialized/in-memory payload before measured file headers and allocator effects.

Metric	Value	Interpretation
Compression vs BF16 repository	6.62x	61.1 GB divided by projected packed bytes.
Compression vs official GPTQ-Int4	1.83x	16.9 GB divided by projected packed bytes.
8K int8 KV cache	0.403 GB	Batch one, K and V, 48 layers, 4 KV heads, head dimension 128.
Total default profile	11.281 GB	Weights + KV + 0.75 GB workspace + 0.90 GB reserve.
Nominal headroom	0.719 GB	Fragile; actual free VRAM may be lower.

7.2 Residual-budget scaling

Residual groups	Expert bpp	Weight GB	Total VRAM GB	Nominal fit
0%	2.157	8.634	10.686	yes
5%	2.211	8.832	10.885	yes
10%	2.266	9.030	11.083	yes
15%	2.321	9.228	11.281	yes
20%	2.375	9.427	11.479	yes
25%	2.430	9.625	11.677	yes
30%	2.485	9.823	11.876	yes

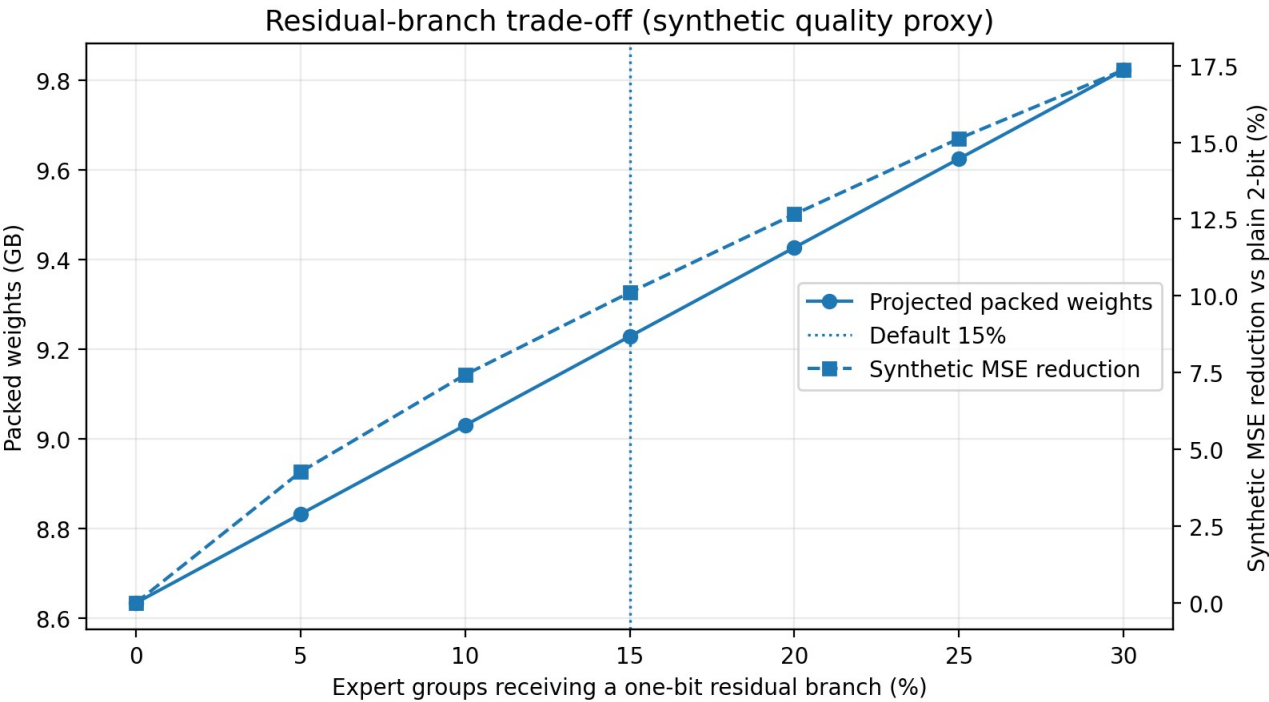


Figure 4. Residual budget increases size while improving a synthetic reconstruction proxy. The synthetic curve is not model accuracy.

7.3 Context scaling

Context	KV GB	Total GB	Headroom GB	Nominal fit
2,048	0.101	10.979	1.021	yes
4,096	0.201	11.080	0.920	yes
8,192	0.403	11.281	0.719	yes
12,288	0.604	11.482	0.518	yes
16,384	0.805	11.684	0.316	yes
32,768	1.611	12.489	-0.489	no

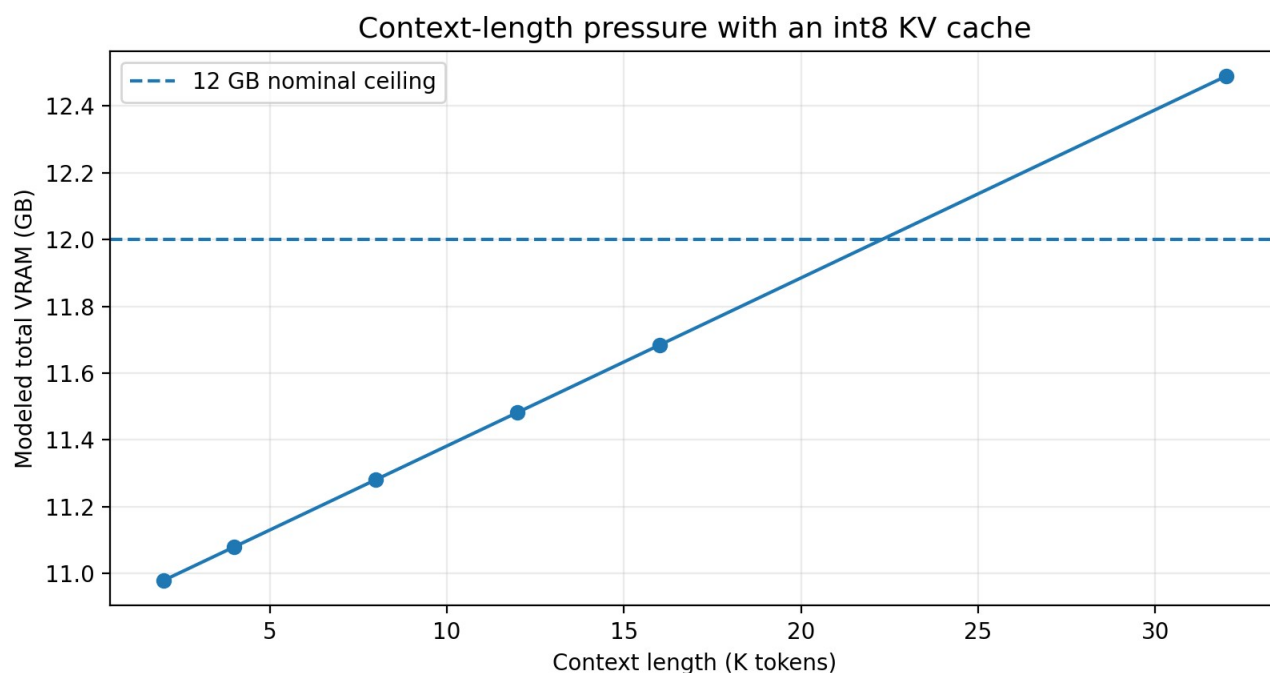


Figure 5. Int8 KV-cache pressure. The 8K default is a deployment choice, not the model's advertised maximum.

8. Execution-speed scaling model

8.1 Active weight traffic

A routed token uses 8/128 of the expert parameters in every MoE layer, while attention, routers and the LM head remain active. Applying the packed bits-per-parameter to those active categories gives 1.182 GB of weight traffic per generated token before cache effects, duplicate reads, activation traffic and kernel metadata. The theoretical memory roofline is therefore $672 / 1.182 = 568$ tokens/s. No general-purpose implementation reaches this roofline.

```

active_expert_params = 28.991B * 8 / 128
active_bytes = active_expert_params * expert_bpp/8
              + (attention + lm_head) * sensitive_bpp/8
              + (router + norms) * 2 bytes
roofline_tps = 672 GB/s / active_bytes

```

8.2 Sensitivity, not forecast

Effective share of published bandwidth	Modeled decode rate	Meaning
2%	11.4 tok/s	Low-efficiency reference or heavy overhead.
3%	17.0 tok/s	Lower engineering sensitivity point.
5%	28.4 tok/s	Mid sensitivity point, not an expected benchmark.
7%	39.8 tok/s	Aggressive but useful optimization target.
10%	56.8 tok/s	Very strong effective packed-weight throughput.

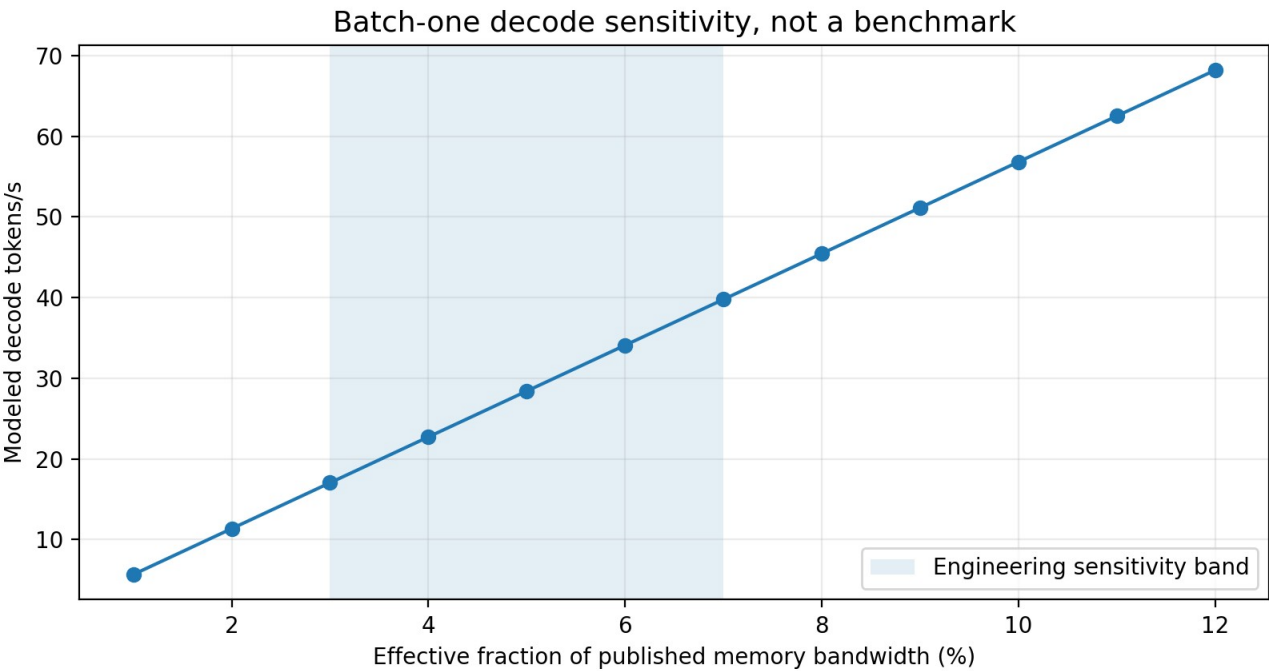


Figure 6. Linear bandwidth sensitivity. Actual kernels may be compute-, launch-, cache-, routing- or power-limited.

8.3 Measurements required

- Kernel median and p10/p90 latency for 2-bit, 4-bit and residual/no-residual shapes.
- Effective bytes read per projection, global-memory transactions, L2 hit rate and achieved bandwidth.
- Time to first token, prefill tokens/s and steady-state decode tokens/s separately.
- Peak allocated and reserved VRAM, including model load and prompt growth.
- Power, clock, temperature and thermal-throttling behavior across the laptop's 60-115 W configurations.
- Five or more post-warm-up runs with fixed prompts and exact software versions.

9. Reconstruction quality, model quality and falsification

9.1 Executed synthetic smoke result

The package executed a deterministic 512 x 1,024 synthetic matrix experiment with low-rank structure, noise and sparse outliers. Relative to plain groupwise 2-bit, the default 15% residual branch reduced reconstruction MSE by 10.1% while maintaining a maximum local zero-set residual of 6.54e-08; stored parity recomputed successfully. The realized in-memory test payload was 2.301 bpp. Four-bit reconstruction had lower MSE, as expected.

Do not over-interpret this result

A synthetic weight matrix does not measure language-model perplexity, task accuracy, routing stability or downstream behavior. It verifies that the codec's correction branch changes the intended error metric and that the local constraint/parity machinery works.

9.2 Minimum model-level comparison

Baseline / ablation	Question
BF16 source	Upper reference for logits, routes and quality.
Official GPTQ-Int4	Existing larger-than-12-GB quality/format reference.
Plain 2-bit groupwise	Does the NHDF residual branch help at its extra memory cost?
Plain 3-bit or equal-size codebook	Can a simpler representation match the same bytes?
NHDF without B0	Does local zero-set mean projection add value beyond an ordinary zero point?
NHDF without log-polar / Delta2 phase	Does the source-derived ranking outperform pure weighted MSE?
Residual fractions 0-30%	Where is the quality/size knee on the exact Qwen checkpoint?
Parity disabled	Does the event gate matter to runtime recovery/telemetry, apart from CRC?

9.3 Required quality metrics

- Held-out perplexity or another declared language-model likelihood metric.
- Instruction, reasoning and code tasks matching intended deployment.
- Long-context retrieval at 4K and 8K.
- Logit cosine similarity, top-k token agreement and greedy output drift.
- Expert-route divergence: fraction of tokens whose top-8 expert set/order differs from BF16.
- Per-layer activation error to locate tensors that need a higher residual budget or precision.

9.4 Fault and resource tests

- Flip one payload bit: parity and CRC should detect it.
- Flip two bits in one parity group: demonstrate parity's blind spot while CRC fails.
- Corrupt/delete a tensor, alter geometry or leave a meta parameter: loader must fail closed.
- Cross residual-mask boundaries at groups 31/32 and expert row boundaries.
- Reduce free VRAM below the projected requirement: doctor/loader must stop before partial activation.
- Run long prompts and repeated generations to expose hidden allocation growth or driver reset risk.

10. Edge applications and operational guide

10.1 Practical edge applications

Application	Why the selected model/architecture is relevant	Primary validation metric
Private local coding assistant	Large sparse instruction model without cloud data transfer; batch-one decode is central.	Code task pass rate, TTFT, decode rate, VRAM.

Application	Why the selected model/architecture is relevant	Primary validation metric
Offline document and RAG analyst	Local reasoning over sensitive corpora; 8K context is a useful initial bound.	Retrieval answer quality, groundedness, 8K stability.
Field-service copilot	Low-connectivity laptop operation; model remains resident between tasks.	Power-limited sustained speed, reliability, domain accuracy.
Research/notebook agent	More capability than smaller dense models may provide while keeping all experts locally stored.	Tool-call accuracy, latency, reproducibility.
Local multi-agent orchestration	One resident model can serve specialized prompts without network dependency.	Concurrent memory, scheduling latency, safety constraints.
Smaller-model efficiency profile	Same codec can trade residual budget for longer context or lower power even when fit is not the problem.	Quality per byte/joule versus standard 4-bit.

10.2 Local workflow

```
# 1. Install and test
python -m pip install -e ".[dev]"
pytest -q
nhdf-edge smoke

# 2. Pull source weights (not included in ZIP)
python scripts/pull_model.py --output models/Qwen3-30B-A3B-Instruct-2507

# 3. Stream-convert
nhdf-edge pack models/Qwen3-30B-A3B-Instruct-2507 packs/qwen3-nhdf \
  --config configs/qwen3_30b_a3b_edge12.yaml

# 4. Verify every file and group parity
nhdf-edge verify packs/qwen3-nhdf --parity-all

# 5. Build target CUDA extension and inspect free VRAM
export TORCH_CUDA_ARCH_LIST="12.0"
python setup_cuda.py build_ext --inplace
nhdf-edge doctor --config configs/qwen3_30b_a3b_edge12.yaml
```

10.3 Benchmark workflow

```
python scripts/benchmark_kernel.py --bits 2 --residual-fraction 0.15
python scripts/benchmark_kernel.py --bits 4 --residual-fraction 0
python scripts/benchmark_model.py packs/qwen3-nhdf --max-new-tokens 64 \
  --output metrics/local/model_benchmark.json
```

The full BF16 teacher does not fit the target GPU. Optional activation calibration therefore needs CPU offload and enough system RAM, or a larger temporary machine. The data-free converter remains usable and should be tested first.

11. Engineering gates, risks and decision criteria

Gate	Exit evidence	Current status
G0 semantic codec	Deterministic CPU tests, local residual tolerance, CRC/parity failure tests.	Included; 41 tests passed in artifact environment.

Gate	Exit evidence	Current status
G1 CUDA equivalence	Build, sanitizer, row-offset tests, numerical error and kernel benchmark JSON.	Source included; target run required.
G2 full conversion	Complete manifest, actual bytes, no missing tensors, policy audit.	Requires local source checkpoint.
G3 model correctness	No meta state, reference logits/routes, deterministic greedy prompts.	Loader included; full pack and CUDA required.
G4 quality/ablations	Predeclared model metrics against equal-budget baselines.	Not run.
G5 laptop performance	TTFT, prefill/decode, VRAM, bandwidth, power and thermals across profiles.	Not run.
G6 prefill optimization	Packed GEMM or bounded dequantize-GEMM within workspace.	Design target.

11.1 Risk register

Risk	Impact	Mitigation / falsifier
Actual pack exceeds projection	No safe 12 GB fit.	Measure serialized categories; lower residual budget, context or choose a smaller model.
2-bit expert quality loss	Model unusable despite fit.	Activation-weighted calibration, per-layer residual budgets, 3-bit/equal-size baseline; reject if thresholds fail.
Router sensitivity	Different experts and cascading logit error.	Keep router FP16; measure route divergence; selectively raise nearby tensors.
Kernel efficiency below useful threshold	Slow generation.	Nsight profile, vectorized loads, expert compaction, optimized packed GEMV/GEMM; compare AQLM/other kernels.
Prefill bottleneck	Poor TTFT on realistic documents.	Layer-local dequantize-GEMM or packed GEMM; measure separately from decode.
VRAM reserve too optimistic	OOM/driver reset.	Use doctor, internal-display tests, configurable context/workspace, fail-before-load.
Thermal throttling	Unstable sustained rate.	Test 60/balanced/115 W profiles with temperature and clock telemetry.
Framework drift	Loader breaks after update.	Pin tested versions, keep adapter isolated, maintain fixed model vectors.
Parity misunderstood as correction	False integrity claims.	CRC/ECC boundary in code/report; explicit two-bit fault test.

11.2 Go/no-go criteria

- GO to optimization only if the complete pack loads with at least the predeclared safe free-VRAM margin and passes CRC/parity/model-structure checks.
- GO to application trials only if model quality meets a declared threshold and NHDF beats a simpler equal-size baseline on at least one relevant metric.

- NO-GO for the 30B target if prefill/decode remains below the application's minimum useful latency after reasonable kernel work; reuse the codec on a smaller model instead.
- NO-GO for any resilience claim unless controlled fault injection demonstrates the specific detection/recovery mechanism.

12. Conclusion

The formal NHDF specification can be translated into a coherent edge-AI codec without relying on physically impossible or unsupported claims. Its strongest implementable idea is the ordered closure of local validity, compact addressing, a bounded event-driven correction branch, causal execution and feedback. In this package, that becomes a mixed-precision MoE weight representation with explicit invariants and a targetable fused decode kernel.

The selected Qwen3 model is a credible challenge: a conventional 61.1 GB BF16 or 16.9 GB official GPTQ-Int4 checkpoint does not fit 12 GB, whereas the default pack is projected at 9.23 GB and the complete 8K profile at 11.28 GB. The fit is narrow but sufficient to justify a controlled experiment. The projected 17.0-39.8 token/s sensitivity band is not evidence of achieved speed; it is a benchmark target tied to an explicit traffic model.

Recommended next action

Run a partial conversion and target CUDA microbenchmark first. Only after CPU/GPU equivalence and measured effective bandwidth should the full 61.1 GB source checkpoint be converted. Then evaluate actual pack bytes, model quality and 8K memory behavior before investing in prefill optimization.

Appendix A. Analytical equations

A.1 Effective expert bits per parameter

```
b_expert = b_base + f
           + 32/g           # FP16 mean + FP16 scale
           + 16f/g + 8f/g   # residual scale + log-polar address
           + 3/g           # mask + prefix + parity
           + 0.02          # header/alignment allowance
```

Default: b_base=2, f=0.15, g=256 -> 2.32078125 bpp

A.2 Packed weights

```
M_weights = N_expert * b_expert/8
           + N_sensitive * b_sensitive/8
           + N_raw * 2 bytes

N_expert = 28,991,029,248
N_sensitive = total - expert - router - norms
N_raw = router + norms
```

A.3 KV cache

```
M_KV = batch * context * 2(K,V) * layers * kv_heads * head_dim * bytes_per_element
At batch=1, context=8192, layers=48, kv_heads=4, head_dim=128, int8 -> 402,653,184 bytes
```


A.4 Decode traffic and roofline

```
active_expert = N_expert * 8/128
B_token = active_expert*b_expert/8 + active_sensitive*b_sensitive/8 + active_raw*2
roofline = published_bandwidth / B_token
modeled_tps(efficiency e) = e * roofline
```

Appendix B. Telemetry and manifest contract

Every model-level run should record enough information to reproduce both numerical and systems behavior. The following is a minimum contract; application adapters may add focus, task or safety residuals.

```
{
  model_repo, model_revision, pack_format, config_hash,
  tensor_count, actual_pack_bytes, crc_failures, parity_failures,
  torch_version, transformers_version, cuda_runtime, driver, device,
  free_vram_before_load, peak_allocated, peak_reserved,
  context, batch, prompt_hash, seed, power_mode,
  ttft, prefill_tps, decode_tps, temperature, power, clock,
  quality_metrics, route_divergence, saturation, overflow, status
}
```

Manifest field	Purpose
format / created_utc / source_model	Version and origin.
config	Complete model, target and packing assumptions.
tensor file / CRC32 / bytes	Integrity and actual storage.
original shape / rows / columns / groups	Decode geometry and bounds.
policy	Raw/quantized, bit width, group size, residual fraction and phase parameters.
stats	Original parameters, effective bpp, residual fraction, rho range, local-field residual and reconstruction error.

Appendix C. Delivered package map

Path	Contents
README.md	Feasibility summary, commands, evidence boundary and references.
configs/	Default Qwen3/12-GB packing profile.
src/nhdf_edge/	Operators, quantizer, pack format, converter, metrics, CLI, calibration and runtime.
csrc/ + setup_cuda.py	Experimental fused packed GEMV and selected-row decode extension.
scripts/	Model pull, calibration, generation, kernel/model benchmarks and report metrics.
tests/	41 CPU tests covering codec, storage, metrics and runtime semantics.
metrics/	Analytical JSON/CSV and deterministic synthetic smoke results.
figures/	Report charts generated from the saved metrics.
docs/	This report, implementation gates and validation protocol.
sources/	NHDF Formal Specification v0.1 and source checksums; no model weights.

References

ID	Source
[S0]	Supplied 30-page prompt-chain transcript: original vocabulary, applications and GPU discussion. Reviewed for traceability; not redistributed in the ZIP because it contains personal identifiers and superseded claims.
[S1]	Tom Klootwijk. Non-Euclidean Holographic Data Fields: A Formal Operator Specification for a Parity-Conditioned Kinematic Foliation and Implicit Holographic Co-Processor. Version 0.1, 31 August 2026. Included in sources/.
[M1]	Qwen Team. Qwen/Qwen3-30B-A3B-Instruct-2507 model card, configuration and checkpoint tree. Hugging Face, accessed 31 August 2026.
[M2]	Hugging Face Transformers. modeling_qwen3_moe.py: official Qwen3MoeExperts tensor layout and forward contract, accessed 31 August 2026.
[M3]	Qwen Team. Qwen3-30B-A3B-GPTQ-Int4 checkpoint tree, 16.9 GB, Hugging Face, accessed 31 August 2026.
[H1]	NVIDIA. GeForce RTX 50 Series Laptop GPU specifications: RTX 5070 Ti Laptop GPU, 12 GB GDDR7, 672 GB/s, 5,888 CUDA cores, 992 AI TOPS, 60-115 W. Accessed 31 August 2026.
[H2]	NVIDIA. CUDA Toolkit 12.8 release notes and Blackwell support documentation.
[Q1]	V. Egiazarian et al. Extreme Compression of Large Language Models via Additive Quantization. arXiv:2401.06118, 2024.
[Q2]	A. Tseng et al. QuIP#: Even Better LLM Quantization with Hadamard Incoherence and Lattice Codebooks. arXiv:2402.04396, 2024.
[Q3]	Y. J. Kim, R. Fahim, H. H. Awadalla. Mixture of Quantized Experts. arXiv:2310.02410, 2023.
[Q4]	Fast Inference of Mixture-of-Experts Language Models with Offloading. arXiv:2312.17238, 2023. Notes that practical 2-bit formats carry metadata and keep embeddings, logits, gates and norms at higher precision.

Citation note

External sources establish model dimensions, checkpoint sizes, hardware specifications and low-bit precedent. None validates the NHDF format, exact Qwen quality or target laptop speed; those remain experimental obligations.

End of report